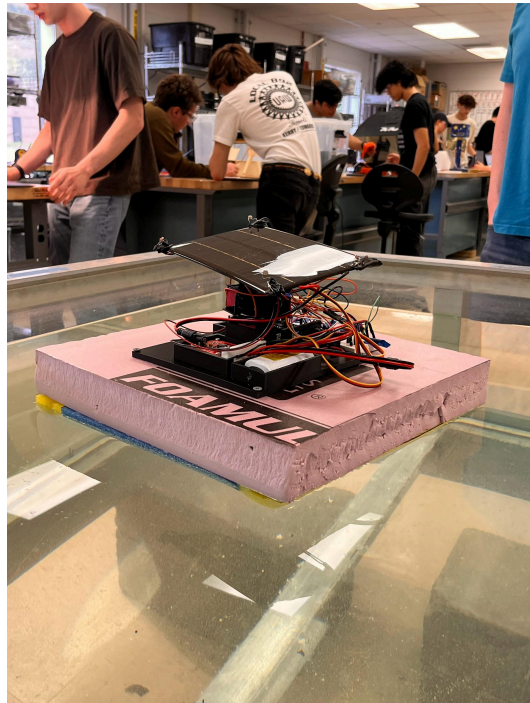


Solar Tracking for Autonomous Drone Boats



April 26, 2026

Prepared by:

The Ohmericans

Kevin Amalraj, Mihir Epel, Curtis Harding, Nathan Trumbull

Engineers in Training

Prepared for:

Professor Theodore Norris, Sarah Burcon, Kristin Drogos

E100 Solar Power Teaching Team

Justin Roelefs, Lab Manager

Jeffrey Shulkin, GSI

Taron Ghlijyan, TA

Table of Contents

Executive Summary.....	<i>iv</i>
Introduction.....	1
Problem Overview.....	1
Current Solutions and Their Limitations.....	1
Why Our Solar Tracking Device is Valuable.....	1
Project Goals and Objectives.....	1
Solution Concept.....	2
Part One: Creating Buoyancy and Waterproofing.....	2
Part Two: Setting Up the Electrical Components.....	2
Part Three: Coding Our Project to Track Light.....	3
Project Requirements.....	3
Requirement 1: The prototype must be able to float.....	3
Requirement 2: The electronics must be protected from water.....	3
Requirement 3: The gimbal tracks the sun through boat movement.....	4
Requirement 4: The boat functions should be powered fully by the sun.....	5
Requirement 5: The tracking device must shut down when light levels are low.....	5
Requirement 6: The use of solar tracking must increase efficiency.....	5
Testing and Results.....	6
Testing Procedure.....	6
Results.....	6
Prototype Analysis.....	9
Requirement 1.....	9
Requirement 2.....	9
Requirement 3.....	9
Requirement 4.....	9
Requirement 5.....	10
Requirement 6.....	10
What Would a Second Design Cycle Look Like?.....	11
Value Proposition.....	11
Brief Stakeholder Analysis.....	11

Cost Analysis.....	12
Recommendation.....	13
Conclusion.....	14
References.....	15

Table of Figures

Figure 1: Initial Sketch of Project Idea.....	2
Figure 2: CAD design of electronics mount.....	4
Figure 3: Light tracking Arduino code.....	4
Figure 4: Reference data showing the difference in efficiency among fixed, single-axis, and dual-axis solar panels.....	5
Figure 5: Raw data for net power measurements.....	7
Figure 6: A graph of the data being aggregated.....	8
Figure 7: A graph of aggregated energy stored without electronics power draw.....	9
Table 1: Cost Estimation for Core Technical Systems Per Unit.....	12

Executive Summary

Marine Biologists spend a lot of time tracking migration patterns of sea life, and a growing number of autonomous drone boats are being used for this. Many of these are now implementing solar power, and a stationary solar panel often lacks the efficiency required. This can lead to incomplete data being collected and a misinterpretation of marine migration patterns.

With this issue, we decided to implement solar tracking in a way that can be utilized by marine biologists. We wanted to create a device that uses photoresistors to read light intensity values and point the solar panel to a spot where it can maximize the amount of light.

In order to make this, we utilized many different items and skills from our time in the lab, such as 3D modeling and coding. The device had to be able to float while operating, and we had to have all of the electronics connected correctly so they could operate together in a beneficial way for us.

To test our project, we used the water tank that is set up in our lab and a lamp to simulate the sun. We tested by moving the light and moving the boat, starting the solar panel in different positions each time.

Our device was successfully able to track the light source in all of our tests, and we found that it generates 40% more power over the course of a day when compared to a stationary solar panel on average.

This report further elaborates on these topics, including how we identified the problem, developed and tested our solution, and why it is an important solution to our problem.

Introduction

This section will provide an overview of the problem being addressed, why it is important, and some of the existing ideas and solutions.

Problem Overview

Many marine biologists who are interested in tracking the migration patterns of different sea species use autonomous drone boats to follow and track the animals, relaying data back to them [1]. In recent years, these scientists have been moving towards solar powered boats, but the problem with these is that sometimes the solar power is not efficient enough. This is caused by waves rocking the boat as well as the sun moving throughout the day, meaning the solar panel is not always pointing to the spot with the highest light intensity. What this can cause is a loss of valuable data that the marine biologist can use to map the migration patterns, leading to inaccurate mapping and a lack of true understanding of the sea life's patterns.

Current Solutions and Their Limitations

There aren't many current solutions being implemented when it comes to solar tracking on autonomous drone boats, but there are currently drone boats out there using solar tracking. Outside of the obvious limitation that has already been addressed with a lack of solar efficiency, there are also legal limitations. Some of these boats get into trouble with laws regarding data collection and maritime boundaries. [2]. These challenges are making it hard for solar power to be integrated into the world of marine biology tracking.

Why Our Solar Tracking Device is Valuable

With an effective solar tracking system, autonomous drone boats would not only be more efficient but would be able to collect much more valuable data. This increase in data collection could result in a discussion of reducing legal limitations on tracking by autonomous boats and further advancing the use of these devices.

Project Goals and Objectives

The goal of our solution is to make a device that can be attached to drone boats that allows for solar tracking to increase their solar efficiency. This involves many different

electronics and setups we learned earlier in the semester, such as servos, arduinos, 3D CAD modeling, and setting up a breadboard to connect all of the electronic parts.

In order to achieve this goal, we had a few key objectives. This included a base that was able to float and keep the electronics safe from water, as well as an electronic setup that allowed for free and correct movement of the solar panel towards the point of maximum light intensity.

Solution Concept

Our solution consists of a raft made from foam and pool noodles, with a 3D printed base for an electronic setup. There were 3 main parts to this solution. We developed an initial sketch, which you can see below, that lays out the main parts of the project.

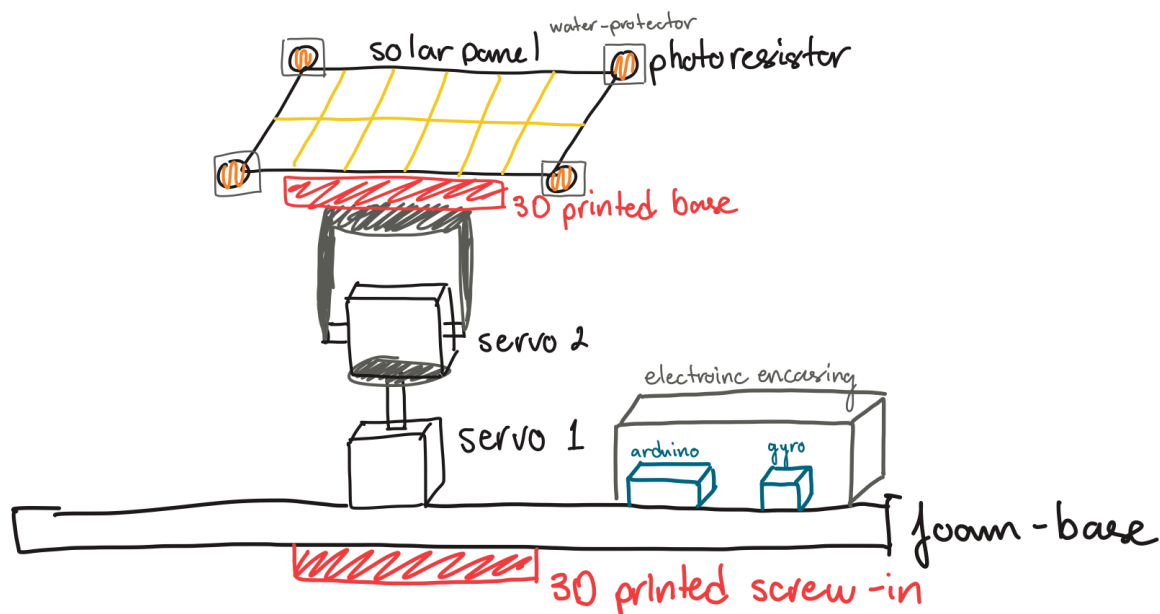


Figure 1: Initial sketch of project idea.

Part one: Creating buoyancy and waterproofing

In this stage, we found materials around the lab that would create a stable, high buoyancy platform for our project. We found some thick, firm foam and pool noodles to construct this. We then went on to create a 3D printed casing for the electrical components with walls on the side that can be placed in the center of the foam.

Part two: Setting up the electrical components

Once we had the base set up, we were able to orient and place all of our components into their set spots. We did most of the planning for this during the creation of the enclosure on CAD, so all we had to do was make sure everything was wired correctly. Once it was fully assembled, the solar panel was attached to two servos that allowed it to rotate on two axes. It had photoresistors on each corner, which read in light intensity values for the Arduino it is connected to, which tells the solar panel where to move.

Part three: Coding our product to track light

The final part of our solution involves a file of code that tells the system what to do based on the values read in by the photoresistors. This code is sent through the Arduino and directs the movement of the 2-servo system.

Project Requirements

To solve the problem of a lack of solar efficiency in preexisting autonomous boats used for tracking marine life, we identified a series of requirements that need to be met when designing our prototype.

Requirement 1: The prototype must be able to float

The prototype is mounted on a thick raft made of foam and pool noodles. A wide piece of foam was used to increase stability and to prevent the boat from tipping due to the movement of the solar panels and the waves.

Requirement 2: The electronics on the boat must be protected from splashing water

The electronics were placed in a 3D printed housing to hold them in place and to protect them from water. The construction of the raft also contributes to this requirement by raising the electronics far from the water, further protecting them. The figure below shows the CAD model for the 3D printed housing.

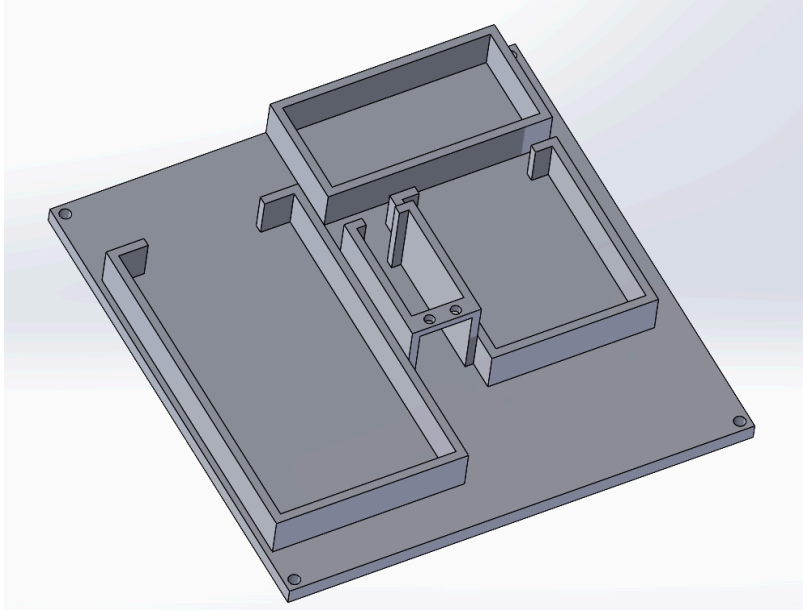


Figure 2: CAD design of electronics mount

Requirement 3: The gimbal must be able to track the sun with the movement of the boat and with the movement of the sun throughout the day

Four photosensors were mounted on the solar panels on each of its corners. These sensors collect light intensity values from the sun, and the Arduino uses this data and the code written below to triangulate the position of the sun and orient the solar panel to face it.

```

37
38 void loop() {
39     float topR = analogRead(A0);
40     float right = analogRead(A1);
41     float left = analogRead(A2);
42     float topL = analogRead(A3);
43
44     Serial.println(left);
45     Serial.println(right);
46     Serial.println(topL);
47     Serial.println(topR);
48
49     topL += 30;
50
51
52
53     // --- PAN CONTROL ---
54     float panError = topL + left - right - topR;
55
56     if (abs(panError) > threshold) {
57         panPos -= (panError > 0) ? -1 : 1;
58     }
59
60     // --- TILT CONTROL ---
61     float avgLR = (left + right) / 2;
62     float top = (topL + topR) / 2;
63     float tiltError = top - avgLR;

```

Figure 3: Light tracking Arduino code

Requirement 4: All of the functions on the autonomous boat are sustained by the power generated from the sun

The power used by the Arduino, Sunny Buddy, photosensors, and servos is less than the power generated by the solar panel. This ensures that the autonomous boat will be self-sustaining and can charge the battery it has onboard.

Requirement 5: The solar tracking device must shut down when light levels become too low

When the set minimum photosensor value is reached, our program tells the gimbal to stop searching for the sun. This allows the boat to save power during the night or in cases when tracking the sun uses more power than what will be generated.

Requirement 6: The benefit of running the solar tracking device must outweigh the benefits of a fixed solar panel

For the gimbal device to be worth the investment, the boat must generate more power than a fixed solar panel model. This is represented by the figure below

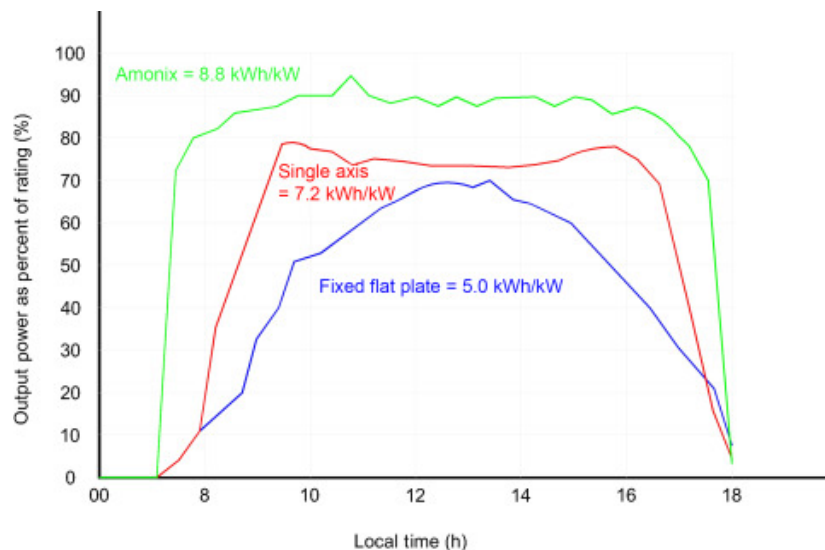


Figure 4: Reference data showing the difference in efficiency between fixed, single axis, and dual axis solar panels

The data collected from the prototype replicates this graph and therefore has a significant benefit over a fixed solar panel mode

Testing and Results

In this section, we will be describing the testing procedure and looking at the data that we collected from our prototype. We will use this to judge the effectiveness of our device.

Testing Procedure

The main goal of our testing was to compare our prototype with a fixed solar panel under lab conditions that replicate a typical day of sunlight. To achieve this, we used a lamp that replicated the light of the sun and measured the net power flowing into the battery at different angles, and used that data to estimate the total energy across the entire test.

In each of the tests, we calibrated the lamp to have an irradiance of approximately 300 Watts per square meter, which is about four times less than the irradiance of the sun, but under the project constraints, it was the closest to a real daytime test in the limited time that we had. To replicate the movement of the sun over a day, we placed the lamp at an angle, because in Ann Arbor, we can expect the sun to be approximately 30 degrees off of straight-up during this time of year [2]. We took measurements starting with the solar panel pointing straight up, and took measurements at incrementing angles, replicating the sun setting. By making the assumption that the time between measurements is irrelevant, which relies on the solar panel not heating up too much, we can make a rough estimate of how much more efficient our device is compared to a fixed solar panel. Because this is a prototype, our main goal was to establish a proof of concept so that we could understand whether our hypothesis was validated and whether the references that we found could be replicated.

Results

The figure below shows the raw data. We don't take data for when the panel is perpendicular to the lamp, we felt that it would mostly take in light from the ambient room lights, and wouldn't accurately reflect sunlight. Thus, the first data point is at 6 degrees. On the vertical axis, we have the net power flowing into the battery. This includes the power draw from the Arduino, the Sunny Buddy, and, for the testing with the gimbal, it includes the draw from the servos and photoresistors.

Net Power Draw/Gain With and Without Gimbal

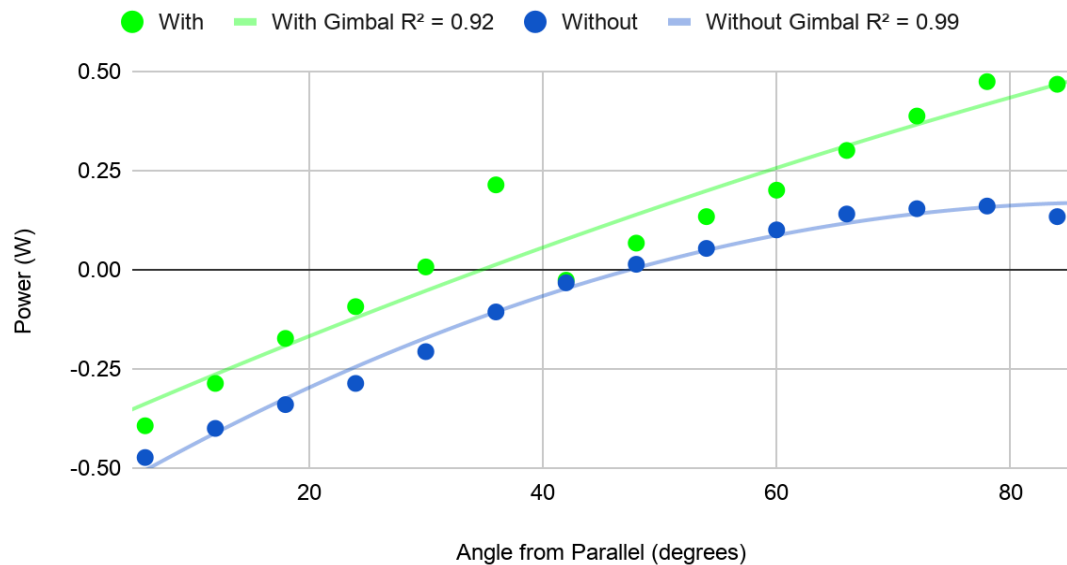


Figure 5: Raw data for net power measurements

The most obvious detail is that most of the power measurements are negative, implying that the battery is being drained instead of charged. This is an expected result for the irradiance of the lamp and the quality of the solar panel itself.

In the next figure below, the data has been aggregated. The units used for the vertical axis are joules, as we assume about one second passes in between measurements. The main purpose of the graph is not to represent the exact charge of the battery, but rather to gain an estimate as to how much more efficient the prototype is.

Aggregate Energy With and Without Gimbal

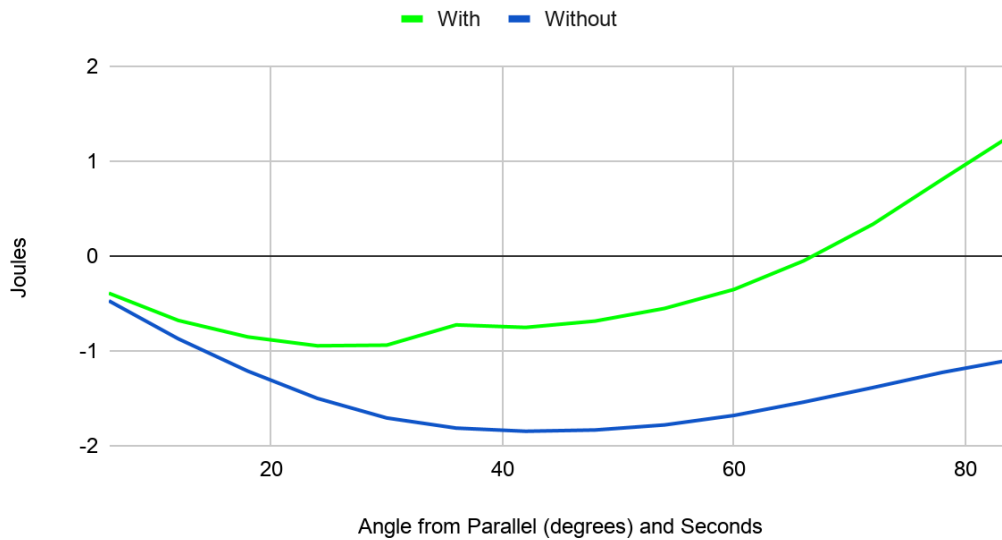


Figure 6: A graph of the data being aggregated

Just like the previous graph, we see negative values. This graph gives us an idea of the energy actually gained by our prototype. We can see that at this lower light level, the fixed solar panel cannot even sustain the minimal electronics.

Finally, the figure below shows the graph above, except without the power draw from the Arduino and Sunny Buddy. The data for the prototype still includes the power draw from the servos and photoresistors to provide an accurate comparison.

Aggregate Energy With and Without Gimbal Without Electronics Power Draw

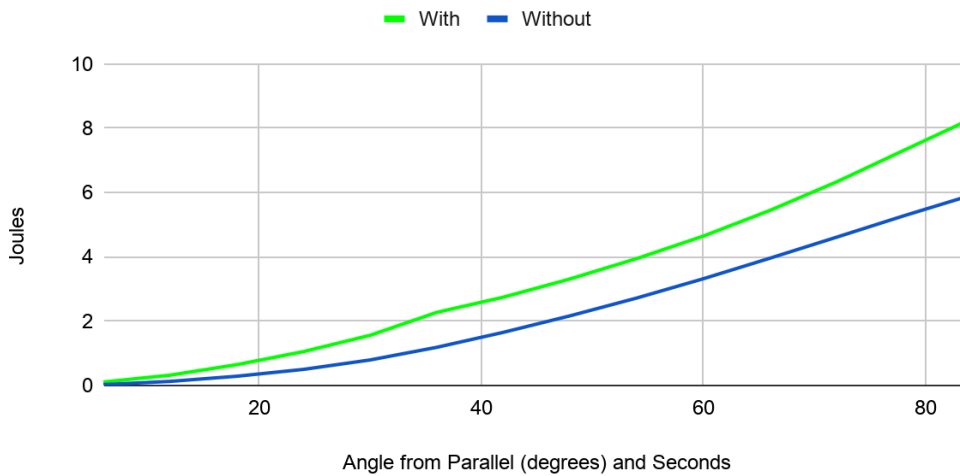


Figure 7: A graph of aggregated energy stored without electronics power draw

From this data, we can conclude that our prototype is around 40% more efficient over the course of a typical day than a fixed solar panel.

Prototype Analysis

From our results, our prototype successfully proves the viability of our idea. That being said, we did not meet all of our requirements to their fullest extent. This section will evaluate each requirement individually.

Requirement 1: The prototype must be able to float

The prototype was able to float, and even with the movement of the solar panel, the prototype was never really in any danger of taking on water or capsizing. This requirement was fulfilled completely.

Requirement 2: The electronics on the boat must be protected from splashing water

In this aspect, the prototype was not entirely up to the standard that we had initially set. Because the foam and pool noodles were significantly more buoyant than required, the electronics sat pretty high, but nearly all electrical components were exposed from

above. Full waterproofing would have required every connection to be soldered and thermally shielded, and the servos to be waterproof.

Requirement 3: The gimbal must be able to track the sun with the movement of the boat and with the movement of the sun throughout the day

We succeeded in accomplishing this requirement. The solar panel can track at a speed of approximately 60 degrees per second, far more than the movement of the sun, and while further testing is needed, we believe that this should be reasonable for the movement of the boat in the waves.

Requirement 4: All of the functions on the autonomous boat are sustained by the power generated from the sun

This requirement fell out of the scope for our prototype. We initially wanted to replicate the functions of existing autonomous boats, like movement and active marine life tracking, but with time and monetary constraints, we realized that those goals weren't feasible. However, the solar panel sustains the current electronics.

Requirement 5: The solar tracking device must shut down when light levels become too low

This is another requirement that fell out of the scope of our project. While the actual implementation seems easily achievable, the ability to test the effects of such a feature wasn't there. We would need to have a continuous test that spanned over days, and we simply didn't have that amount of time.

Requirement 6: The benefit of running the solar tracking device must outweigh the benefits of a fixed solar panel

This was probably the most important requirement for our prototype. In the end, a 40% gain in efficiency is extremely significant even if you account for any possible systematic error. Not only is the gain in efficiency significant, but we believe that it is far beyond the threshold to be useful in a commercial product.

What Would a Second Design Cycle Look Like?

If we had the opportunity to go back and create a second prototype, we would be able to focus less on the solar tracking system by taking inspiration from the first prototype, and we would focus more on testing real world conditions, like placing the prototype in a lake or pond. This would require us to implement shutting down at night and better waterproofing, but would greatly increase the quality of our data and provide concrete evidence of our product's viability. We might also try to add the tracking functionality to our prototype, which would prepare it for a real tracking test. With these improvements, we would be able to start designing for the open ocean.

Value Proposition

Overall, we highly advise investing in this product because it will provide significant advancement towards autonomous marine technology and the widespread adoption of solar-powered vehicles

Brief Stakeholder Analysis

The primary stakeholders being targeted are the marine research institutions and environmental monitoring agencies, since their tasks would benefit greatly from the operation of autonomous boats. These tasks would include marine life tracking, water quality monitoring, and hydrographic mapping. The secondary stakeholders would include defense contractors, offshore energy companies, and private academic research institutions, as they too would require data collection in areas that are remote and only safely accessible through autonomous vehicles such as our prototype.

Our prototype's capabilities will provide these stakeholders with a more cost-effective and energy-efficient option to the standard gas-powered engine boats. We will be offering the dual-axis gimbal solar-tracking system in two different applications. The first is where the system will be fully integrated into a newly built autonomous boat, allowing for maximum efficiency for the hardware and software components of the vehicle. The second will be a standalone attachable system that can be easily integrated into existing gas-powered vessels. Although this seems as if it stems away from our primary purpose, this hybrid integration would encourage full solar-efficiency implementation to reluctant stakeholders who don't want to fully invest in the complete construction of the autonomous boat. The existence of these two package options

would provide a higher likelihood for the stakeholder to invest in the product and even come back after more development has occurred.

Cost Analysis

The core expenditures that this product requires in order for it to be launched on a full scale, based on the design of the prototype, are the capital expenditures, which include the hardware infrastructure needed to build and deploy a unit, and operating expenditures, which include the costs for maintenance and operation of the system.

The capital expenditures for these prototypes are primarily centered on the five necessary technical systems that must be effectively functional for the autonomous boat to operate as designed. These systems include the dual-axis gimbal, marine-grade solar panels, energy storage, the boat's base hull, and the boat's propulsion. The estimated cost range for each component will be outlined in Table 1 below. Also, it is important to note that the vehicle would also require the data processing and collection hardware systems, since the cost of these systems would vary significantly based on the type of data processing and collection. We will provide a reasonable estimation of the infrastructure's cost.

Table 1: Cost Estimation of Technical Systems Per Unit

Technical System	Cost
Dual-Axis Gimbal	\$20,000-\$30,000
Marine Grade Solar Panels	\$1,500
Marine Grade Battery	\$1,500-\$2,500
Hull (full integration)	\$5,000
Propulsion	\$800
Data Processing	\$100-\$200
Data Collection	\$500-\$10,000

Given the table above, we can conclude that the lower bound cost of a single fully designed unit is approximately \$29400, and the upper bound cost would be approximately \$45500.

Recommendation

We recommend investing in this product for three core reasons. Its modular design, its superior energy efficiency, and its broad adaptability beyond the current intended application.

The first reason why we recommend investing in the product is its modular design. The system is designed so that all the components are easily attachable and removable from their initial positioning without the need for time-intensive disassembly processes that could also risk damaging the components. A common real-world occurrence in the constant usage of boats is the need to frequently have the vehicle undergo maintenance, repair, and improvements. None of that can be easily done without the ability to swap and move around the component modules. In addition, reduced maintenance time results in a reduction of labor costs and increased availability for workers to spend time on other pressing tasks.

The second reason for our recommendation of the product is its improved energy efficiency relative to the boats with standard fixed-axis solar panels. The core benefit of the dual-axis gimbal is its ability to orient itself towards the sun. This maximizes the amount of energy captured, as a panel not facing directly towards the sun doesn't have the ability to absorb as much of the sun's light. This becomes especially important if the solar panel is placed on a moving vehicle situated in a disruptive environment, as there are more factors that could make a fixed-axis solar panel not face the sun. However, the dual-axis solar panel will hunt for the sun using the metrics necessary, and improve the energy input. Improved energy input will keep the electrical components powered for longer periods of time, alongside allowing the autonomous boat to have increased performance.

The third feature that is worth investment is its adaptability. Although the current prototype is designed for marine life tracking, its core innovation, the dual-axis gimbal tracking system, can be implemented in any scenario where a fixed solar panel wouldn't perform as well. These other systems include especially autonomous land-fixed vehicles, unmanned aerial vehicles, and stationary objects such as buoys. This dynamic implementation nature allows this product to grow beyond its current intended audience and, in turn, produce a larger margin of profit. It is important, however, to keep in mind that scaling this prototype to full commercial deployment will present some challenges that require further analysis towards its

appropriate solution. The boat's navigation in harsh climates is still a notable concern, as anything severe could result in partial or full damage to the boat. In addition, several logistics on when and why the boat would need to return to the initial station would need to be sorted out so that we can minimize risks to the boat. But, nonetheless, we believe that these challenges are not insurmountable and that with further investments towards the overall growth of this product, we can mainstream the long-term vision of introducing renewable energy usage in gas-powered markets.

Conclusion

The viability of solar powered autonomous boats has yet to be solved, in large part due to the loss of solar efficiency from the movement of the sun throughout the day and waves. We have sought to solve this problem by developing a 2-axis gimbal to track the sun. Our objectives for this project were to create a prototype that could track the sun, as well as generate more power than a fixed solar panel model, and be able to sustain all of the boat's functions when there is little to no sunlight.

By testing our prototype in a two-axis configuration and a fixed configuration, we found that our two axis gimbal generated significantly more power than the fixed solar panel model. Additionally, we measured and calculated the power usage of our device and found that it used less power than it generated, allowing it to be self-sustaining. Lastly, by testing our prototype in the water, we proved that it was able to float, and the waterproofing used to protect the electronics was effective.

To improve upon our initial prototype, we plan to design and test for harsher conditions that better match the ocean, where our commercial product will eventually be operated. To accomplish this, we will implement a gyroscope to help counteract the movement of the boat, waterproof the electronics further to improve reliability, and implement measures to improve power efficiency at night. To test our prototype in more accurate conditions, we will use a pond or a lake, which have harsher conditions than the controlled tank we used initially.

References

- [1]J. Markoff, “No Sailors Needed: Robot Sailboats Scour the Oceans for Data,” *The New York Times*, Sep. 05, 2016. Available: <https://www.nytimes.com/2016/09/05/technology/no-sailors-needed-robot-sailboats-scour-the-oceans-for-data.html>
- [2]Aspasia Pastra, T. M. Johansson, J. Soares, and F. E. Muller-Karger, “The use of emerging autonomous technologies for ocean monitoring: insights and legal challenges,” *Frontiers in Marine Science*, vol. 12, Aug. 2025, doi: <https://doi.org/10.3389/fmars.2025.1561737>.
- [3]Geotimedate.org, 2026. <https://geotimedate.org/sun/usa/mi/ann-arbor> (accessed Apr. 24, 2026).
- [4]C. Lane, “What Is A Solar Tracker And Is It Worth The Investment?,” *Solarreviews.com*, Apr. 24, 2024. <https://www.solarreviews.com/blog/are-solar-axis-trackers-worth-the-additional-investment#are-solar-trackers-worth-the-additional-investment>? (accessed Apr. 24, 2026).
- [5]“Highest Efficiency (Up to 25.4%) Rigid Marine Solar Panels For Your Boat in 2025. We’ll help you decide on the right panels to comfortably meet your boat’s power needs,” *Premium Marine Solar Panels, Complete Solar Systems, and Lithium Iron Batteries*, 2025. <https://www.custommarineproducts.com/rigid-panel-specifications.html> (accessed Apr. 24, 2026).
- [6]“Lithium Marine Batteries For Boats and Vans - 2025 Lithium Battery Technology And Bluetooth To Phone App And Removable BMS,” *Premium Marine Solar Panels, Complete Solar Systems, and Lithium Iron Batteries*, 2025. <https://www.custommarineproducts.com/lifepo4-lithium-batteries.html> (accessed Apr. 24, 2026).
- [7]“BlueBoat Archives,” *Blue Robotics*, 2025. <https://bluerobotics.com/product-category/boat/> (accessed Apr. 24, 2026).
- [8]“T500 Thruster,” *Blue Robotics*. <https://bluerobotics.com/store/thrusters/t100-t200-thrusters/t500-thruster/>
- [9]“Controllers,” *Holybro Store*, 2025. <https://holybro.com/collections/flight-controllers> (accessed Apr. 24, 2026).
- [10]E. Upton, “16GB Raspberry Pi 5 on sale now at \$120 - Raspberry Pi,” *Raspberry Pi*, Jan. 09, 2025. <https://www.raspberrypi.com/news/16gb-raspberry-pi-5-on-sale-now-at-120/>

[11]A. Shukla, R. Ross, B. Bhattacharya, and A. Stumpf, "Autonomous water sampling and quality monitoring in remote locations: A novel approach using a remote-controlled boat," *HardwareX*, vol. 22, p. e00634, Jun. 2025, doi: <https://doi.org/10.1016/j.ohx.2025.e00634>